Instructions for Running the Assignment 1 Solution

Group 05

1 Overview

This document explains how to execute the Assignment 1 solution for chi-square based term selection using MapReduce with Python and `mrjob`.

The project contains three MapReduce jobs:

- `chi_square_job_1.py` — preprocessing, token filtering, and review/category counting
- `chi_square_job_2.py` — aggregation of intermediate statistics
- `chi_square_job_3.py` — chi-square computation and top-term selection

A formatting script, `format_output.py`, is used to generate the final required `output.txt` file.

2 Project Structure

The expected folder structure is:

```
Group05_DIC2026_Assignment_1/
|
|-- src/
| |-- chi_square_job_1.py
| |-- chi_square_job_2.py
| |-- chi_square_job_3.py
| |-- format_output.py
| |-- stopwords.txt
|
|-- data/
| |-- reviews_devset.json
|
|-- run_all.sh
|-- output.txt
|-- README.md
|-- instructions.pdf
|-- report.pdf
```

3 Requirements

Before running the project, make sure the following are installed:

- Python 3
- `pip`
- `mrjob`

To install `mrjob`, run:

```
python -m pip install mrjob
```

4 Input Files

The solution requires:

- a review dataset in JSON-lines format
- a stopword file

By default, the script uses:

```
data/reviews_devset.json
src/stopwords.txt
```

5 How to Run the Full Pipeline

The complete solution can be executed with a single command:

```
bash run_all.sh
```

This script performs the following steps automatically:

1. Runs Job 1 and creates `job1_output.txt`
2. Runs Job 2 and creates `job2_output.txt`
3. Runs Job 3 and creates `final_output.txt`
4. Runs the formatter and creates the final `output.txt`

After successful execution, the terminal should display:

```
Running Job 1...
Running Job 2...
Running Job 3...
Formatting final output...
Done. Final result written to output.txt
```

6 Using a Different Input File

The shell script also supports custom input and custom stopwords file.

General format:

```
bash run_all.sh <input_file> <stopwords_file>
```

Example:

```
bash run_all.sh data/reviews_devset.json src/stopwords.txt
```

7 Intermediate Output Files

During execution, the following intermediate files are created:

- `job1_output.txt`
- `job2_output.txt`
- `final_output.txt`

These are intermediate results used for debugging and verification. The final submission result is stored in:

```
output.txt
```

8 Description of Final Output

The final `output.txt` file contains:

- one line for each category
- the top 75 terms for that category in descending order of chi-square score
- one final line containing the merged dictionary of selected terms in alphabetical order

Each category line has the format:

```
<category> term1:score1 term2:score2 ... term75:score75
```

The last line contains:

```
termA termB termC ...
```

9 Notes

- All paths in the implementation are relative.
- The stopword file is loaded through the `-stopwords` argument in Job 1.
- Tokenization, lowercasing, stopword removal, and one-character filtering are handled in preprocessing.
- Chi-square is computed in Job 3 using the aggregated statistics from the previous jobs.

10 Troubleshooting

If the command

```
bash run_all.sh
```

does not work in Windows Command Prompt, run it in Git Bash.

If `mrjob` is missing, install it again using:

```
python -m pip install mrjob
```

11 Conclusion

The provided shell script offers a complete end-to-end execution of the solution, starting from the input dataset and stopword file and ending with the final `output.txt` file required for submission.